

REVISION HISTORY

Version No.	Date	Originator	Summary of Changes
1	07-10-2025	C.E.N, A.R, E.D, A.M.R	Document Created
2	04-11-2024	E.D	Modified Design Constraints, reframed some requirements
3	25-03-2026	E.D	Updated Design Constraints, updated detection thresholds
4	07-04-2026	C.E.N, A.R	Updated requirements, constraints and key requirements

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Requirement Specification

This section outlines the functional and performance requirements of the system, detailing the design objectives, operational constraints, and expected behaviors necessary to ensure that the implementation meets its intended purpose.

Functional Requirements:

The functional requirements for the design and implementation of this project are listed below:

Functionality (ID: 01):

Type of Constraint	Requirement Description
Performance/Capability	The robot shall be able to follow the user within a 100 cm radius when not in standby.
Performance/Capability	The robot shall be able to follow the user at a speed of 5 km/h (± 1 km/h) when following a user.
Performance/Capability	The robot shall be able to carry a payload of up to 25 kg securely during operation.
Performance / Timeliness	The system shall be able to detect and avoid obstacles within 150 cm and reroute to continue following the user.

Hardware (ID: 02):

Type of Constraint	Requirement Description
Performance/Capacity	The system shall be able to accommodate objects not exceeding 50 cm $\times$ 40 cm $\times$ 25 cm in dimension.
Performance/ Operability	The system shall operate on indoor floor surfaces with coefficient of friction ≥ 0.5 and ≤ 0.9
Performance/ Operability	The system shall operate on indoor floor surfaces with maximum surface unevenness ≤ 5 mm
Performance/ Operability	The system shall operate on indoor floor surfaces with maximum incline $\leq 5^\circ$
Performance/ Operability	The system shall operate on indoor floor surfaces with no debris larger than 2 cm

UI (ID: 03):

Type of Constraint	Requirement Description
Functionality / Accessibility	The system shall be able to provide a user interface to start and stop operation.

Power (ID: 04):

Type of Constraint	Requirement Description
Performance / Timeliness	The system should be rechargeable.
Sustainability / Power	The system shall be able to recharge using standard 120 V AC power and reach full charge within 6 hours.
Performance / Power	The system shall be able to last for 4 hours on a single charge cycle under 25 kg load

Non-Functional Requirements:

The non-functional requirements for the design and implementation of this project are listed below:

Environmental (ID: 05):

Type of Constraint	Requirement Description
Environmental / Operability	The system shall be able to function within enclosed spaces and shall not be exposed to outdoor conditions such as rain, direct sunlight, or uneven terrain.
Environmental / Operability	The system shall be able to operate indoors only, within room temperature (15–30 °C).
Environmental / Operability	The system shall be able to function in an under environmental conditions with noise levels of up to 50dB.

Security (ID: 06):

Type of Constraint	Requirement Description
Security / Data integrity	The system shall not store any visual/audio data of the user.
Security / Access Control	The system shall be able to pair and operate only with authorized user devices or identifiers.

Design Constraints

The constraints below define the limitations and conditions that influence the system's design:

1. The system shall be designed for indoor operation only under controlled environmental conditions such as stable temperatures and dry surfaces.
2. The system shall be designed to support navigation only on smooth, flat surfaces to ensure traction and stability.
3. The system shall be limited to operating on a single floor level, without the ability to traverse stairs, ramps, or steep inclines.
4. The system shall be capable of carrying a payload of not more than 1kg while maintaining stable motion on smooth surfaces.
5. The system shall be developed within a total budget not exceeding \$1000 CAD.
6. The system design shall accommodate a maximum object size of 50cm x 40cm x 25cm.
7. The system shall be battery-powered and rechargeable using a standard electrical outlet.

APPENDIX I: KEY REQUIREMENTS

Requirement #	R01-01
Requirement Type:	Functional
Description:	The robot shall be able to follow the user within a 100m radius when not in standby.
Originator:	Client (G5)
Fit Criterion:	The robot maintains a following distance within a 100m radius from the user during motion, verified through testing in an indoor environment.
Priority:	1
Conflicts:	None
History:	Date Created: 8 th October, 2025 Adjusted for Final Design: 7 th April, 2026

Requirement #	R01-02
Requirement Type:	Functional
Description:	The robot shall be able to carry a payload of up to 25 kg securely during operation.
Originator:	Client (G5)
Fit Criterion:	The robot shall successfully transport a payload of 25 kg without loss of balance, slippage, or system malfunction over a continuous operation period on a smooth indoor surface.
Priority:	2
Conflicts:	None
History:	Date Created: 8 th October, 2025 Adjusted for Final Design: 7 th April, 2026

Requirement #	R01-03
Requirement Type:	Functional
Description:	The system shall be able to detect and avoid obstacles within not less than 150cm and reroute to continue following the user.
Originator:	Client (G5)
Fit Criterion:	During testing, the system consistently detects obstacles at a minimum distance of 40 cm and successfully adjusts its path to avoid collision while maintaining continuous user tracking.
Priority:	3
Conflicts:	None
History:	Date Created: 9 th October, 2025 Adjusted for Final Design: 7 th April, 2026

Requirement #	R02-01
Requirement Type:	Hardware
Description:	The system shall operate on indoor floor surfaces with coefficient of friction ≥ 0.5 and ≤ 0.9
Originator:	Client (G5)
Fit Criterion:	The system shall demonstrate stable motion, including forward movement, turning, and stopping, on indoor surfaces with a coefficient of friction between 0.5 and 0.9, without wheel slippage or loss of control during operation.
Priority:	4
Conflicts:	None
History:	Date Created: 9 th October, 2025 Adjusted for Final Design: 7 th April, 2026

Requirement #	R02-02
Requirement Type:	Hardware
Description:	The system shall operate on indoor floor surfaces with maximum surface unevenness ≤ 5 mm
Originator:	Client (G5)
Fit Criterion:	The system shall successfully traverse indoor surfaces with surface unevenness up to 5 mm over a continuous path of at least 5 m, without loss of stability, tipping, or degradation in tracking performance.
Priority:	5
Conflicts:	None
History:	Date Created: 7 th April, 2026

Requirement #	R02-03
Requirement Type:	Hardware
Description:	The system shall operate on indoor floor surfaces with maximum incline $\leq 5^\circ$
Originator:	Client (G5)
Fit Criterion:	The system shall demonstrate continuous operation on an inclined surface of up to 5° , maintaining controlled movement, stable speed, and accurate user-following behavior without rollback or instability.
Priority:	6
Conflicts:	None
History:	Date Created: 7 th April, 2026